



Prenatal and postnatal exposure to di-(2-ethylhexyl) phthalate and neurodevelopmental outcomes: A systematic review and meta-analysis

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ABSTRACT

Background: Di-(2-ethylhexyl) phthalate (DEHP), the most widely used phthalate, has recently been associated with neurodevelopmental disturbances in children. However, the risk is yet to be quantified. Therefore, a systematic review and meta-analysis focusing on the association between exposure to DEHP and neurodevelopmental outcomes is necessary, with particular attention to study design (longitudinal vs. cross-sectional).

Methods: We performed a comprehensive literature search for associations between exposure to DEHP and neurodevelopmental outcomes. Among 106 published studies found in public databases, eight longitudinal studies and two cross-sectional studies were included in the meta-analysis.

Results: We observed a statistically significant association between the concentrations of DEHP metabolites and the neurodevelopment outcomes of children among cross-sectional results, and found significant association between DEHP exposure measured in prenatal period and the psychomotor development outcomes measured later in childhood.

Conclusions: To our knowledge, this is the first meta-analysis of studies investigating the association between DEHP exposure and neurodevelopment in children. A need exists for more researches and a precautionary policy for potential health hazard of DEHP, the most commonly used phthalate, to promote healthier neurodevelopment in children.

1. Introduction

Phthalates are widely used as plasticizers to promote flexibility and softness of plastics and in various personal care products. In the United States, urinary phthalate metabolites were detected in more than 75% of Americans in 1999–2000 (Silva et al., 2004). Phthalates are a class of chemicals synthesized by esterification of phthalic acid. The estimated consumption of di-(2-ethylhexyl) phthalate (DEHP), the most widely used phthalate, is 65.2% (Lorz et al., 2000) of the total consumption of phthalates, which can be as high as 325 million tons per year.

Phthalates are well-known endocrine disrupting chemicals; therefore, studies on phthalate health effects have focused on abnormal sexual development, adverse birth outcomes, and hormonal

disturbances (Borch et al., 2006; Kim et al., 2009; Lee et al., 2014; Moore et al., 2001; Swan, 2008). Recently, phthalate exposure has been associated with neurodevelopmental abnormalities in children (Cho et al., 2010; Kim et al., 2011). Ejaredar et al. performed a systematic literature review for phthalate exposure and neurodevelopment in children and suggested that exposure to phthalates induces adverse neurodevelopmental outcomes, including lowered intelligence quotient (IQ). However, quantification of phthalate effects on neurodevelopment in children by synthesis of results across many published studies has yet to be performed.

There are several reasons for inconclusiveness of the association between phthalate exposure and their neurotoxicity. The chemicals classified as “phthalate” and the types of metabolites that were

Abbreviations: BSID, Bayley Scales of Infant Development; DEHP, di-(2-ethylhexyl) phthalate; GABA, gamma-aminobutyric acid; MDI, mental developmental index; PDI, psychomotor developmental index; VIQ, verbal intelligence quotient; VCI, verbal comprehension index; WMI, working memory index; PIQ, performance IQ; PRI, perceptual reasoning index; PSI, process speed index; NOS, Newcastle-Ottawa quality assessment scale; PPAR, peroxisome proliferator-activated receptor; WISC, Wechsler intelligence scale for children

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analyzed to assess exposure are very diverse (Ejaredar et al., 2015). The timing of measurement of phthalate and the methods of assessing neurodevelopment differ across studies as well. Therefore, to synthesize such results by meta-analysis, we targeted the association between DEHP, the most widely used phthalate, and neurodevelopmental outcomes measured by intelligence-assessment instruments validated for infants and children. Our analyses were run separately for longitudinal studies and cross-sectional studies which differed in the timing of measurement for phthalate exposure.

2. Materials and methods

2.1. Search strategy and selection methods

We searched the MEDLINE, EMBASE, Cochrane Library, and PsycINFO databases for research articles that reported an association between DEHP and neurodevelopment between January 1, 1980 and October 31, 2017, using the search string (“phthalates” OR “phthalate” OR “DEHP” OR “diethylhexyl”) AND (“neurodevelopment” OR “neuropsychology” OR “intellectual” OR “intelligence quotient” OR “IQ” OR “mental”).

Inclusion criteria were: (1) cohort, case-control, or cross-sectional design; (2) reported effect estimates, such as beta estimates (β), with 95% confidence interval (CI), or outcome values that allowed for effect to be estimated by calculations in a 2×2 cell table; (3) neurodevelopmental outcomes measured by validated tools such as the Bayley scales of infant development (BSID) or the Wechsler intelligence scale for children (WISC). Exclusion criteria were (1) reported irrelevant outcomes; (2) absence of effect estimates or an inability to calculate the size of the association; (3) letter, commentary, or review article; (4) study population identical to that of a later study (we included only the latest publication after a review of the full text); (5) not in English; and (6) nonhuman subjects.

Two authors (DL and MK) searched the literature and selected researches according to inclusion criteria. They excluded animal studies, in vivo studies, and studies in research field other than health or medicine in the title level, and mainly excluded letter, commentary, or review article, and studies not in English after assessing abstracts. They then reviewed the full texts to determine eligibility according to the inclusion/exclusion criteria. Disagreements regarding eligibility were resolved via discussion and agreement with a third author (YH). Finally, they manually checked the reference lists of the articles identified by the automated search for further leads.

2.2. Data extraction

The following information was extracted from all included studies: the surname of the first author, year of publication, study design, country, study population, the timing of measurement of phthalate metabolites, the metabolites of DEHP measured, the methods of statistical analysis, the adjustment variables, the instrument used for neurodevelopmental assessment, and the beta estimates with 95% CIs. The measured DEHP metabolites varied across studies, and results of analyses were presented separately according to each DEHP metabolite measured in each of the included studies, with or without the sum of DEHP metabolites. We preferentially selected the sum of DEHP secondary metabolites as an exposure indicator. Among the secondary metabolites, we selected the available metabolites in the order: mono-(2-ethyl-5-carboxypentyl) phthalate, mono-(2-ethyl-5-hydroxy-hexyl) phthalate (MEHHP), and mono-(2-ethyl-5-oxo-hexyl) phthalate, considering the molar fraction of excretion to absorbed DEHP in the human body (Koch et al., 2005). When any of the secondary metabolites of DEHP were not available, we used mono-(2-ethylhexyl) phthalate as an indicator of exposure. When beta estimates with 95% CIs were presented for more than one model, the association calculated with the greatest number of adjustment variables was used for meta-analysis.

2.3. Quality assessment

We used the Newcastle-Ottawa quality assessment scale (NOS) (Wells et al., 2013). The NOS for cohort and cross-sectional studies consists of items for selection, comparability, and outcome assessment. Using the NOS, cohort studies were scored in the range 0–9 and classified as low-quality (0–3), moderate-quality (4–6), or high-quality (7–9). Cross-sectional studies were scored in the range 0–10, and similarly classified as low-quality (0–3), moderate-quality (4–7), or high-quality (8–10). Studies of low quality were originally to be excluded from the analyses by agreement among three authors (DL, MK, and YH), but there was no such study among those reviewed at the full-text level.

2.4. Statistical analysis

Two main meta-analyses were performed independently according to the design of the study, longitudinal vs. cross-sectional. In the result of systematic review for the association between phthalates and neurodevelopment outcome, all included studies used WISC and BSID to assess neurodevelopment outcome. We investigated the pooled association size between phthalates and neurodevelopment outcomes as the full-scale intelligence quotient (FSIQ) of WISC and the mental development index (MDI) of BSID according to the design of the study. These indices both measure cognitive function, have a population mean of 100 and a standard deviation of 15 (Chen et al., 2014). Furthermore, we investigated the pooled association size of phthalates on components of WISC and components of BSID, among longitudinal studies vs. cross-sectional studies.

The heterogeneity of results across studies was examined by Q test. A p value of less than 0.10 was interpreted as substantial heterogeneity. When this was observed, the overall estimate was calculated using a random-effects model. Otherwise, a fixed-effects model was used. In the final round of analysis, we evaluated publication bias of the included studies using a Begg funnel plot and the Egger test. If the funnel plot was asymmetric or the p value from the Egger test was less than 0.05, we assumed the existence of publication bias, the possibility that a study without significant result was not published. Statistical analyses were conducted using R software, “metaphor” package, version 2.13.2 (Wolfgang Viechtbauer, Maastricht, the Netherlands).

3. Results

Fig. 1 shows the process used to identify relevant studies. Of the 106

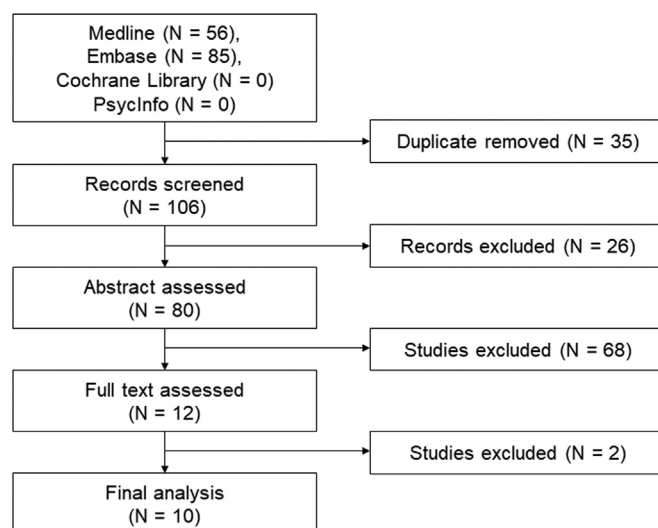


Fig. 1. Flow diagram of the study selection process.

Table 1
Summary of observational studies included in meta-analyses. All point estimates were β estimates.

No	First Author	Year	Study Design	Country	Study Population	Timing of Measurement	Measured DEHP Metabolites	Range of Metabolites	Statistical Model Type	Adjustment Variables	Instrument for Neurodevelopmental Assessment
1	Huang HB	2015	Panel	Taiwan	11 y, $n = 251$	Prenatal maternal urine, Child's urine	GM of MEHP + MEHHP + MEOHP	58.69 $\mu\text{g/g}$ crea (maternal); 184.55 $\mu\text{g/g}$ crea (child, 2–3 y); 167.61 $\mu\text{g/g}$ crea (child, 5–6 y); 89.46 $\mu\text{g/g}$ crea (child, 8–9 y); 72.11 $\mu\text{g/g}$ crea (child, 11–12 y)	Linear mixed	Gender, HOME score, birth weight, maternal education, lactation, and child's age.	BSID-II (MDI), WPPSI-R, WISC-III, WISC-IV
2	Huang PC	2017	Cross-sectional	Taiwan	3–6 y, $n = 108$; 6–12 y, $n = 96$	Child's urine	MEHP + MEHHP + MEOHP	4.4–701.4 $\mu\text{g/g}$ crea (3–6 y); 23.8–704.4 $\mu\text{g/g}$ crea (6–12 y)	Multiple regression	Age, gender, maternal IQ, education, passive smoking, estimated past DEHP exposure, active and passive smoking during pregnancy.	WPPSI-R (3–6 y), WISC-IV (6–12 y)
3	Factor-Litvak P	2014	Cohort	USA	7 y, $n = 328$	Prenatal maternal urine	MEHHP	1.1–1750 $\mu\text{g/g}$ crea	Multiple regression	Specific gravity, maternal IQ, ethnicity, alcohol use during pregnancy, education, marital status, total HOME score, and gender of child.	WISC-IV
4	Kim JI	2017	Cohort	Korea	6 y, $n = 175$	Prenatal maternal urine, Child's urine	GM of MEHHP	12.5 $\mu\text{g/g}$ crea (maternal); 64.1 $\mu\text{g/g}$ crea (child)	Multiple regression	Maternal education level, maternal full-scale IQ, and maternal CPT.	K-WISC-III
5	Cho SC	2010	Cross-sectional	Korea	9 y, $n = 667$	Child's urine	MEHP, MEOHP	0.506–445.376 $\mu\text{g/g}$ crea; MEOHP, 0.065–291.111 $\mu\text{g/g}$ crea	Multiple regression	Age, gender, birth weight, history of breast-feeding, residential area, paternal education, SES, and maternal IQ.	KEDI-WISC
6	Polanska K	2014	Cohort	Poland	2 y, $n = 165$	Prenatal maternal urine, Child's urine	MEHP + MEHHP + MEOHP	0.0004–15 $\mu\text{mol/g}$ crea (maternal); 0.0004–2.2 $\mu\text{mol/g}$ (child)	Multiple regression	Examiner, parental age, parental education, child gender, pre- and post-natal ETS exposure; (additionally for cognitive development) marital status and child nursery attendance.	BSID-III
7	Doherty BT	2016	Cohort	USA	2 y, $n = 247$	Prenatal maternal urine	GM of MEHP + MEHHP + MEOHP + MECPP	0.28 $\mu\text{mol/L}$; (GM of MEHP, 6.2 $\mu\text{g/L}$; GM of MEHHP, 20 $\mu\text{g/L}$; GM of MEOHP, 18 $\mu\text{g/L}$; GM of MECPP, 35 $\mu\text{g/L}$)	Multiple regression	Urine creatinine, pre-pregnancy body mass index, maternal race, maternal education, HOME score, duration of breastfeeding in months, maternal age in years, child age at testing in months, child's gender, and maternal marital status.	BSID-II
8	Tellez-Rojo M	2013	Cohort	Mexico	3 y, $n = 135$	Prenatal maternal urine	GM of MEHP + MEHHP + MEOHP + MECPP	0.35 nmol/mL	Linear regression	Mother's age, mother's education level, birth weight, breastfeeding practices, Z-score for weight-for-age, and child's current age.	BSID-II

(continued on next page)

Table 1 (continued)

No	First Author	Year	Study Design	Country	Study Population	Timing of Measurement	Measured DEHP Metabolites	Range of Metabolites	Statistical Model Type	Adjustment Variables	Instrument for Neurodevelopmental Assessment
9	Kim Y	2011	Cohort	Korea	6 months <i>n</i> = 227	Prenatal maternal urine	GMs of MEHHP, MEOHP	GM of MEHHP, 8.9 µg/L; GM of MEOHP, 7.4 µg/L	Multiple regression	Infant birth weight, infant gender, maternal age, maternal education level, family income, breast-feeding status, and residential area.	BSID-II
10	Minatoya M	2016	Cohort	Japan	18 months <i>n</i> = 97	Prenatal maternal blood	Median, IQR of MEHP	Median, 10.6 ng/mL; IQR, 6.3–17.1 ng/mL	Multiple regression	Gestational age, mother's age at delivery, maternal education, and blood sampling for MEHP measurement.	BSID-II

y, years of age; DEHP, di-(2-ethylhexyl) phthalate; GM, geometric mean; IQR, interquartile range; MEHP, mono-(2-ethylhexyl) phthalate; MEHHP, mono-(2-ethyl-5-hydroxy-hexyl) phthalate; MEOHP, mono-(2-ethyl-5-oxo-hexyl) phthalate; MECPP, mono-(2-ethyl-5-carboxypentyl) phthalate; crea, creatinine; HOME, home observation for measurement of the environment inventory; BSID, Bayley scales of infant development; WPPSI-R, Wechsler preschool and primary scale of intelligence-revised; WISC, Wechsler intelligence scale for children; KEDI-WISC, Korean Educational Development Institute-Wechsler intelligence scale for children; CPT, continuous performance test; SES, socioeconomic status; ETS, environmental tobacco smoke; K-WISC-III, Korean Wechsler intelligence scale for children-III.

records screened, we excluded 26 studies for relevance of titles of studies. After abstract assessed, 68 irrelevant studies were excluded. After we assessed full text of 12 studies, ten studies met our inclusion criteria. Finally, we checked the reference lists of included studies, and studies not included in the final studies were not found. There were six studies that met inclusion criteria, but all of these 6 studies were already found by the online search.

Table 1 summarizes the ten observational studies, eight longitudinal and two cross-sectional (Cho et al., 2010; Doherty et al., 2017; Factor-Litvak et al., 2014; Huang et al., 2015, 2017; Kim et al., 2017, 2011; Polanska et al., 2014; Tellez-Rojo et al., 2013; Whyatt et al., 2012). The study size varied between 484 and 50,032 participants. Studies were conducted in Korea (*n* = 3), the USA (*n* = 2), Taiwan (*n* = 2), Japan (*n* = 1), Mexico (*n* = 1), and Poland (*n* = 1). The quality of the studies as assessed by the Newcastle-Ottawa scale is given in Table A1-A2. The scores of the included cohort studies (*n* = 8) ranged from 7 to 8, and the included cross-sectional studies (*n* = 2) both scored at 8. No case-control study fulfilled the inclusion criteria. Among eight longitudinal studies, 3 studies were measured DEHP metabolites from both maternal urine during prenatal period and children urine when the neurodevelopment tests were administered. Other five longitudinal studies only measured DEHP metabolites in prenatal maternal urine. Two cross-sectional studies measured only children urine when neurodevelopment tests were performed. Neurodevelopment outcome of the children were assessed at 6 months to 11 years after birth, according to each study. All included studies assessed urinary DEHP metabolites, from either mothers or children, and generated beta estimates by analysing the association between the DEHP concentration and the neurodevelopmental outcome. All studies presented the results as a beta estimate with a 95% CI, using a linear regression model or a linear mixed model. The WISC (*n* = 5) and the BSID (*n* = 5) were used for assessing neurodevelopment and no studies using other assessment tools for assessing neurodevelopment outcome were found. Urinary DEHP-metabolite results were presented using the natural logarithm or the base-10 logarithm of concentration, so that all results were standardized by conversion to the natural logarithm as required.

Fig. 2 shows the results of the meta-analysis of longitudinal studies investigated the association between DEHP exposure and neurodevelopmental outcome. A fixed-effects model was used because heterogeneity among these studies was not significant ($p = 0.504$). No significant pooled effect was found ($\beta = -0.14$, 95% CI = -0.70 to 0.41). The funnel plot shows no deviation (Figure A.1), and publication bias was not shown by the Egger test ($p = 0.677$).

Fig. 3 shows the results of the meta-analysis of cross-sectional results (*n* = 6). Although three studies entered into this analysis employed a cohort design, the associations between the urinary DEHP metabolites of children and neurodevelopment outcomes obtained at the same time were cross-sectionally analyzed (Huang et al., 2015; Kim et al., 2017; Polanska et al., 2014). In one study, neurodevelopment outcomes were presented separately at 3–6 years and at 6–12 years. We included the respective associations in the meta-analysis as two separate results (Huang et al., 2017). Because we found no significant heterogeneity among the included studies ($p = 0.223$), a fixed-effects model was used. The summary β was shown to be significant and negative for the association between DEHP metabolites in children and neurodevelopment outcomes ($\beta = -1.03$, 95% CI = -1.88 to -0.18). Visual inspection of the funnel plot revealed no asymmetry (Figure A.1). The Egger test showed no publication bias ($p = 0.560$).

Finally, we performed meta-analyses separately for pooled association size of phthalates on components of WISC and components of BSID, among longitudinal studies vs. cross-sectional. Fig. 4 shows the result of meta-analyses for five longitudinal studies using BSID. We found no significant association between DEHP and MDI among longitudinal studies ($\beta = -0.36$, 95% CI = -1.05 to 0.32). There was significant association between DEHP metabolites in prenatal maternal urine and psychomotor development index (PDI) of their children using

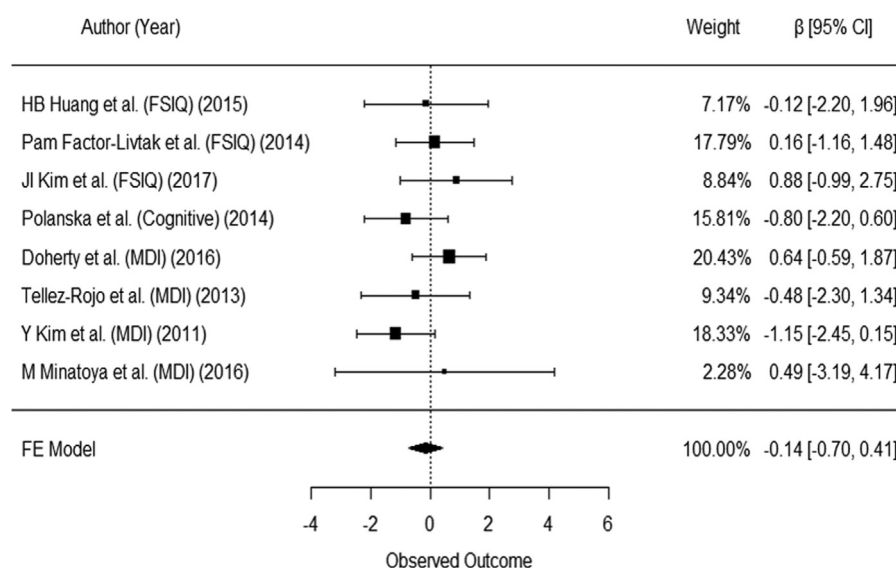


Fig. 2. Forest plot for studies of the association of DEHP exposure with neurodevelopmental outcome: longitudinal studies. DEHP, di-(2-ethylhexyl) phthalate; WISC, Wechsler intelligence scale for children; BSID, Bayley scales of infant development; MDI, mental development index (a BSID subscale); FE, fixed effects.

a fixed-effects model ($\beta = -0.80$, 95% CI = -1.48 to -0.12). The funnel plot (Figure A.1) and the egger test showed no publication bias ($p = 0.491$).

Fig. 5 shows the results of the meta-analysis of cross-sectional associations between DEHP metabolites and the components of WISC. Only three studies using WISC reported the cross-sectional association sizes for components of WISC. In the case of the size of the association was presented verbal IQ as its subindices (verbal comprehension index and working memory index) and performance IQ as its subindices (perceptual reasoning index and process speed index), the results of the association between neurodevelopment outcome and subindices were included separately. The results show that DEHP metabolites in children urine is neither associated with verbal intelligence quotient (VIQ) nor associated with performance intelligence quotient (PIQ). Longitudinal studies with results for components of WISC ($n = 2$) and cross-sectional results for components of BSID ($n = 1$) were not analyzed for pooled association size.

4. Discussion

To our knowledge, this is the first meta-analysis of studies investigating the association between DEHP exposure and

neurodevelopment in children. We found a significant negative association between childhood exposure to environmental DEHP and neurodevelopment outcome measured using the WISC. A 2-fold increase in DEHP metabolites of childhood is associated with a 0.8-point reduction in IQ. Furthermore, a 2-fold increase in DEHP metabolites of prenatal maternal urine is associated with a 0.6-point reduction in PDI of their childhood.

Assessing exposures to DEHP may be not conclusive since various metabolites of DEHP are rapidly metabolized in vivo and quickly excreted. The excretion half-lives of metabolites of DEHP can be as little as 0.5–3.0 days (Koch et al., 2005). Therefore, only recent exposure to DEHP is accurately reflected in urine biomarkers of DEHP. However, all studies included in meta-analyses measured DEHP metabolites of maternal urine or children urine. In all longitudinal studies, DEHP metabolites measured only once from pregnant urine and/or children's urine. Only one study measured children's urine repeatedly 4 times with 2–3 years interval. In all cross-sectional studies, DEHP metabolites measured only once from children's urine.

As its short half-lives and excretion, it has been studied that how metabolites of DEHP have temporal stability over periods of weeks to months. Fromme et al. investigated daily variation of phthalate metabolites with collecting 8 consecutive days of 50 participants, and

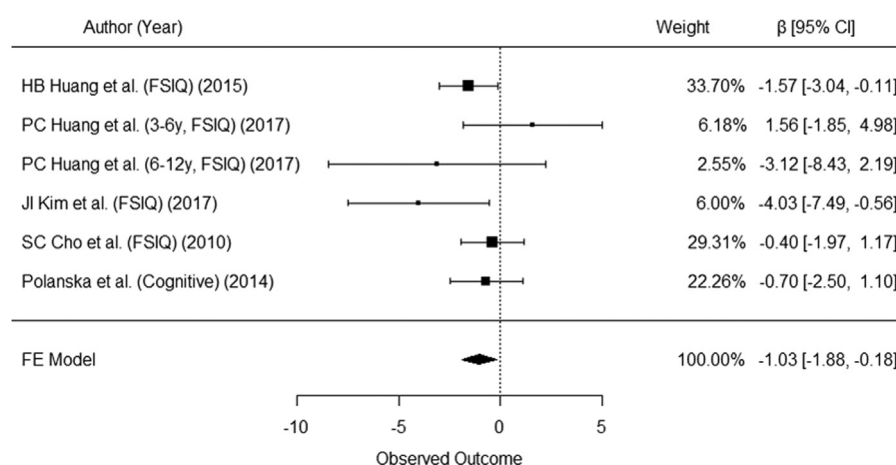


Fig. 3. Forest plot for studies of the association of DEHP exposure with neurodevelopmental outcome: cross-sectional studies. WISC, Wechsler intelligence scale for children; FSIQ, full-scale intelligence quotient; FE, fixed effects.

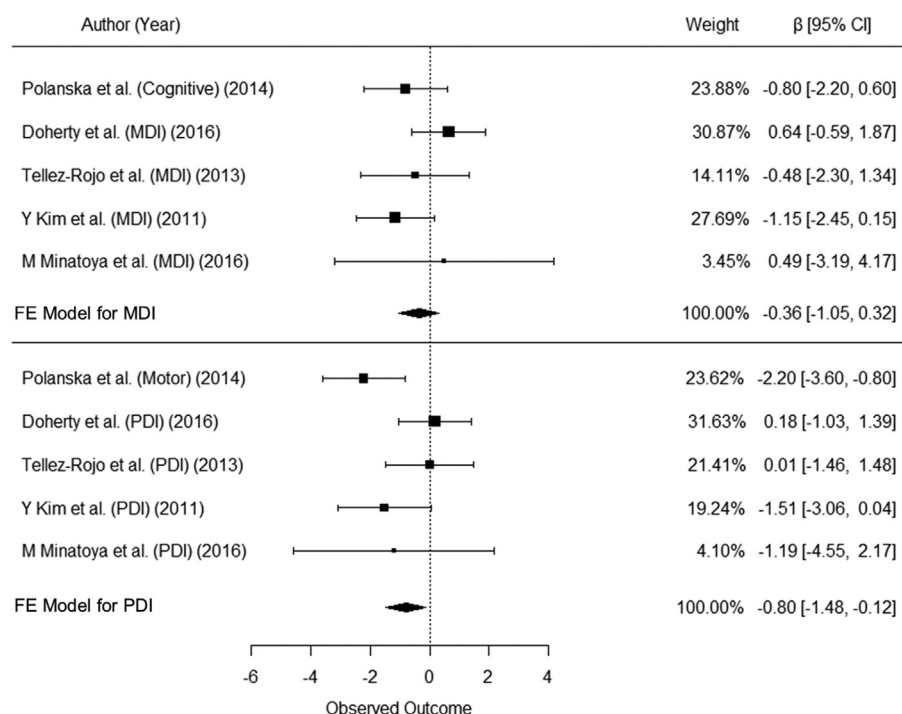


Fig. 4. Forest plot for studies of the association of DEHP exposure with components of Bayley scales of infant development: longitudinal studies. DEHP, di-(2-ethylhexyl) phthalate; BSID, Bayley scales of infant development; MDI, mental development index (a BSID subscale); PDI, psychomotor developmental index (a BSID subscale); FE, fixed effects.

reported intra-class coefficients (ICCs) of creatinine-adjusted DEHP metabolites were 0.20–0.34. (Fromme et al., 2007) Hauser et al. reported that DEHP metabolites in one spot urine sample could predict 3-month average DEHP metabolites levels with sensitivity of 0.56 and specificity of 0.83. (Hauser et al., 2004) DEHP metabolites in spot urine of pregnant women and children could have not excellent stability, but reasonable temporal stability among weeks to months (Adibi et al., 2008; Hines et al., 2009; Hoppin et al., 2002; Suzuki et al., 2009; Teitelbaum et al., 2008). With this background, all studies included meta-analyses assumed implicitly or explicitly that a single measurement could reflect exposure over a considerable period.

We performed meta-analyses and found that increased DEHP

metabolites and decreased neurodevelopment outcome are associated in childhood, and that increased DEHP metabolites in urine from pregnant women are associated with decreased psychomotor development of their childhood. Some longitudinal studies included meta-analyses interpreted their results as prenatal exposure to phthalate could decrease neurodevelopment outcome (Kim et al., 2011; Polanska et al., 2014; Whyatt et al., 2012). Two studies measured DEHP metabolites both in prenatal maternal urine and postnatal children urine at the timing of IQ measurement interpreted their result that postnatal exposure to phthalate, not but prenatal exposure, could decrease neurodevelopment outcome (Huang et al., 2015; Kim et al., 2017). As it is possible that children who have higher phthalate levels are more likely

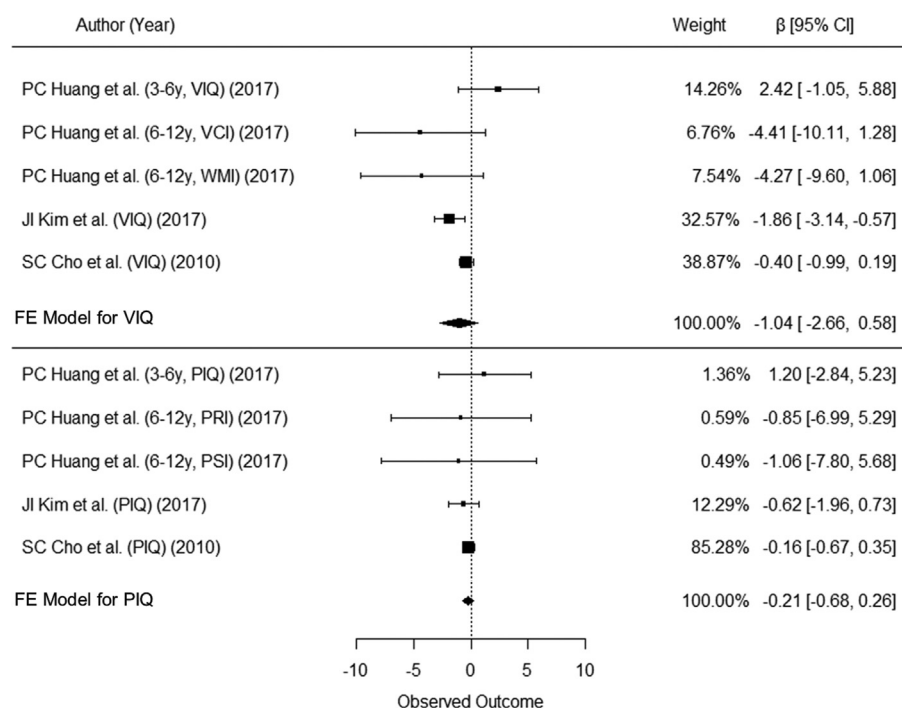


Fig. 5. Forest plot for the analyses with cross-sectional results of included studies for the association of DEHP exposure with components of WISC. WISC, Wechsler intelligence scale for children; FE, fixed effects; VIQ, verbal intelligence quotient; VCI, verbal comprehension index; WMI, working memory index; PIQ, performance IQ; PRI, perceptual reasoning index; PSI, process speed index; RE, random effects; FE, fixed effects.

to be exposed to phthalate prenatally, our results could not conclude that there is more susceptible period of phthalate effects on neurodevelopment disturbance. Our results could be interpreted strictly as exposure to DEHP is associated with neurodevelopment disturbance.

Proposed mechanism for the effects of DEHP on neurodevelopment includes disruption of thyroid homeostasis, myelination process, neurotransmitter system, lipid metabolism, calcium signalling, androgenic effects of DEHP, oxidative stress, and activation of the peroxisome proliferator-activated receptor (PPAR) (Ferguson et al., 2011; Miodovnik et al., 2014).

Disturbances of thyroid-hormone homeostasis are also associated with phthalate exposure (Boas et al., 2010; Wu et al., 2017). Thyroid hormones are important for the neurodevelopmental process in both the prenatal and postnatal periods. Maturation processes in the brain, such as the migration of granule and pyramidal cells in the cortex, are partly regulated by thyroid hormone (Porterfield and Hendrich, 1993; Williams, 2008).

DEHP has the potential to affect the myelination process through its effects on lipid metabolism and distribution. Fetal rats from DEHP-fed dams show an inhibition of cholesterol synthesis in the brain and liver, and a reduction in the whole-body lipid content (Bell, 1982; Xu et al., 2007). Decrease of Gamma-aminobutyric acid (GABA), an important inhibitory transmitter in the mammalian brain during critical periods in neurodevelopment, (Hensch, 2005) has been reported after DEHP exposure of rats in the perinatal period after birth (Carbone et al., 2012, 2010). The balance between synaptic GABA and glutamate release influences connectivity changes and synaptic pruning may be influenced (Howard et al., 2014; Seybold et al., 2012). Dopamine, a neuromodulator associated with cognition, learning, and memory, may be down regulated by exposure to DEHP (Ishido et al., 2005). Calcium signalling, which plays a role in the neurodevelopmental process, in neuron migration, and in the development of neurotransmitter phenotypes, (Komuro and Rakic, 1996; Tang et al., 2003), was suppressed by DEHP in a human neuroblastoma cell line and disturbed intracellular calcium concentrations (Kaun-Yu et al., 2004; Palleschi et al., 2009).

Androgenic effects of DEHP was suggested in some studies included in meta-analyses. Although direct mechanism support for this relationship is not clear, phthalates was suggested as anti-androgenic effects and the importance of sex hormones in brain development (Borch et al., 2006; Colborn, 2004; Gore et al., 2014).

Oxidative stress, a possible chemically induced neuronal damage process, was increased after DEHP exposure in astrocytes from *Caenorhabditis elegans* (Tseng et al., 2013; Wu et al., 2014). The nuclear transcription factors of PPAR, having different distributions depending on the developmental period and regulating neurodevelopment, (Abbott, 2009; Wada et al., 2006) is disturbed by DEHP (Lampen et al., 2003; Lin et al., 2011), although these proposed mechanisms have not yet been confirmed in humans (Xu et al., 2013).

Studies above mentioned did not specify the differences between prenatal and postnatal exposure of DEHP. In this paper, meta-analysis showed significant dissimilarity between two periods. It may be explained by the fact that cognitive function is not developmentally complete at the time of birth (Mark H. Johnson, 2005). Advancing myelination increases the speed of transmission of neuronal action potentials during the postnatal period. Some studies suggested that postnatal exposure to phthalates could be associated with neurodevelopment outcome (Huang et al., 2015; Kim et al., 2017). DEHP exposure may affect the development of the hippocampus during the postnatal developmental period, when rapid structural and functional changes are taking place (Smith et al., 2011). Until the clear evidence on how DEHP influences cognitive development, the relationship between DEHP and neuronal maturing process is still remained to be speculative.

Our study has several limitations. Numerous factors are associated with determining intelligence and confirmation of causal relationship between pre/post-natal exposure to phthalate and neurotoxicity presents various challenges. First, calibration of the amount of exposure to

phthalate with consideration to duration of exposure is practically impossible for now. The fetus is affected through maternal blood or inhalation and the infant is exposed through breast milk as well as environment (Williams and Ross, 2007). The blood-brain barrier is imperfect during fetal and infantile development and other environmental confounding toxins affecting neurodevelopment should be controlled in further studies to confirm the neurotoxic effect of phthalates. Phthalates are also found in many medical devices such as catheters, medical tubing, and blood bags, which are used both prenatally and postnatally, and may add another confounding variables regarding neurotoxicity of phthalate in the context of prenatal and postnatal differences (Ünürar and Büyükgebiz, 2012).

Another difficulty is how best to sum up the different metabolites to assess total exposure amount. The molar sum of several metabolites of DEHP is considered by far the best estimate of exposure rather than a simple mass sum of DEHP metabolites. Therefore, we aimed to use the molar sum of multiple metabolites to estimate the exposure amount. However, some studies used a single metabolite rather than a sum of metabolites. Moreover, the phthalate metabolites measured, and the summation methods used varied across studies. Although in each study we selected the association between the most reasonable metabolites of DEHP and the neurodevelopment outcome, the interpretation of our results should be cautious because the different metabolites may not have the same values for estimating exposure.

In addition, all studies included in the meta-analysis used multivariate linear regression for investigating the association between exposure to DEHP and neurodevelopmental outcomes. For assessing neurodevelopment outcome, various tools were used depending on the age of children when neurodevelopment outcome is measured. The BSID is a tool for measuring development, targeting for children aged 1–42 months. For measuring cognitive and motor function, this tool was commonly used for children younger than 30 months (Bayley, 2006; Limbos and Joyce, 2011). Wechsler preschool and primary scale of intelligence-revised is generally used tool for cognitive function for children older than 30 months of age (Wechsler et al., 2012). BSID and WISC have similar purpose that assessing neurodevelopment of children, but quantified values of these tools have yet been studied for interpretation at the same time. Differences in measuring tools for neurodevelopment across the studies can lead to the potential bias. Furthermore, the adjusting variables differed from study to study. As Table 1 shows, although most studies adjusted for major potential confounders such as age, gender, mother's educational level, maternal IQ, smoking status and score of home observation for measurement of the environment inventory, there were individual differences in methodology and minor items in the context of adjusting variables for each study. Other risk factors for neurotoxicity, such as poverty, poor living conditions, child abuse, and parental attitudes toward child rearing, should be individually analyzed and compared. These issues were discussed in several articles for our meta-analysis, but were insufficient to elucidate all the confounding variables in relation to intellectual development. Although there are various methods of meta-analysis given the difference in the variables used for the regression analyses among included studies, this study used beta estimates and 95% CIs of DEHP, the main independent variable, as a limitation of information and methodological differences among included studies (Becker and Wu, 2007). If raw data can be obtained and pooled analysis carried out, it is expected that more robust results will be obtained.

In this meta-analysis, we found methodological limitations due to measuring intervals, duration of longitudinal cohort studies, and quantitative and qualitative analysis on differences among tests for cognitive development. Prospective cohort studies of large population with repeated measures of mental and psychomotor development with both urine and serum concentration of phthalates over longer periods of time, taking in to account environmental and individual factors, will give more accurate trajectories of neurotoxic effects of phthalates. Future studies should consider better experimental design with

appropriately controlled confounding factors, refined measurement methods of biological exposure indices, and timely assessment of neurodevelopment. Research on the mechanisms of phthalate impairing neurodevelopment is needed to elucidate clear causality (Rauh and Margolis, 2016).

5. Conclusions

The result of this meta-analysis emphasizes that exposure to DEHP carries risks of disturbed neurodevelopment in children. The severity of the problem is not negligible since DEHP is still the most commonly used phthalate. More researches and precautionary policies to caution for health hazards of DEHP should be made to protect neurodevelopment of children.

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DeclarationsEthics approval and consent to participate

Not applicable.Consent for publication

Not applicable.Availability of data and material

All data analyzed during this study are from previously published articles.Authors' contributions

Conceived and designed the study: Dong-Wook Lee, Yun-Chul Hong, Nami Lee

Performing a systematic review: Dong-Wook Lee, Min-Seok Kim, Yun-Chul Hong

Analyzing data and preparing the tables and figures: Dong-Wook Lee, Youn-Hee Lim

Wrote the paper: Dong-Wook Lee, Nami Lee, Yun-Chul Hong

Critically revised the paper: Nami Lee, Yun-Chul HongConflict of interest

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.envres.2018.08.023.

References

- Abbott, B.D., 2009. Review of the expression of peroxisome proliferator-activated receptors alpha (PPAR α), beta (PPAR β), and gamma (PPAR γ) in rodent and human development. *Reprod. Toxicol.* 27, 246–257.
- Adibi, J.J., et al., 2008. Characterization of phthalate exposure among pregnant women assessed by repeat air and urine samples. *Environ. Health Perspect.* 116, 467.
- Bayley, N., 2006. Bayley Scales of Infant and Toddler Development: Bayley-III. Harcourt Assessment, Psych. Corporation.
- Becker, B.J., Wu, M.-J., 2007. The synthesis of regression slopes in meta-analysis. *Stat. Sci.* 414–429.
- Bell, F.P., 1982. Effects of phthalate esters on lipid metabolism in various tissues, cells and organelles in mammals. *Environ. Health Perspect.* 45, 41.
- Boas, M., et al., 2010. Childhood exposure to phthalates: associations with thyroid function, insulin-like growth factor I, and growth. *Environ. Health Perspect.* 118, 1458.
- Borch, J., et al., 2006. Mechanisms underlying the anti-androgenic effects of diethylhexyl phthalate in fetal rat testis. *Toxicology* 223, 144–155.
- Carbone, S., et al., 2010. Impact of gestational and lactational phthalate exposure on hypothalamic content of amino acid neurotransmitters and FSH secretion in peripubertal male rats. *Neurotoxicology* 31, 747–751.
- Carbone, S., et al., 2012. Different effects by sex on hypothalamic-pituitary axis of prepubertal offspring rats produced by in utero and lactational exposure to di-(2-ethylhexyl) phthalate (DEHP). *Neurotoxicology* 33, 78–84.
- Chen, A., et al., 2014. Prenatal polybrominated diphenyl ether exposures and neurodevelopment in US children through 5 years of age: the HOME study. *Environ. Health Perspect.* 122, 856.
- Cho, S.C., et al., 2010. Relationship between environmental phthalate exposure and the intelligence of school-age children. *Environ. Health Perspect.* 118, 1027–1032.
- Colborn, T., 2004. Neurodevelopment and endocrine disruption. *Environ. Health Perspect.* 112, 944.
- Doherty, B.T., et al., 2017. Prenatal phthalate biomarker concentrations and performance on the Bayley Scales of Infant Development-II in a population of young urban children. *Environ. Res.* 152, 51–58.
- Ejaredar, M., et al., 2015. Phthalate exposure and children's neurodevelopment: a systematic review. *Environ. Res.* 142, 51–60.
- Factor-Litvak, P., et al., 2014. Persistent associations between maternal prenatal exposure to phthalates on child IQ at age 7 years. *PLoS One* 9, e114003.
- Ferguson, K.K., et al., 2011. Urinary phthalate metabolites in relation to biomarkers of inflammation and oxidative stress: NHANES 1999–2006. *Environ. Res.* 111, 718–726.
- Fromme, H., et al., 2007. Occurrence and daily variation of phthalate metabolites in the urine of an adult population. *Int. J. Hyg. Environ. Health* 210, 21–33.
- Gore, A.C., et al., 2014. Implications of prenatal steroid perturbations for neurodevelopment, behavior, and autism. *Endocr. Rev.* 35, 961–991.
- Hauser, R., et al., 2004. Temporal variability of urinary phthalate metabolite levels in men of reproductive age. *Environ. Health Perspect.* 112, 1734.
- Hensch, T.K., 2005. Critical period plasticity in local cortical circuits. *Nat. Rev. Neurosci.* 6, 877–888.
- Hines, E.P., et al., 2009. Concentrations of phthalate metabolites in milk, urine, saliva, and serum of lactating North Carolina women. *Environ. Health Perspect.* 117, 86.
- Hoppin, J.A., et al., 2002. Reproducibility of urinary phthalate metabolites in first morning urine samples. *Environ. Health Perspect.* 110, 515.
- Howard, M.A., et al., 2014. Bidirectional homeostatic plasticity induced by interneuron cell death and transplantation in vivo. *Proc. Natl. Acad. Sci.* 111, 492–497.
- Huang, H.B., et al., 2015. Fetal and childhood exposure to phthalate diesters and cognitive function in children up to 12 years of age: Taiwanese Maternal and Infant Cohort Study. *PLoS One* 10, e0131910.
- Huang, P.C., et al., 2017. Intellectual evaluation of children exposed to phthalate-tainted products after the 2011 Taiwan phthalate episode. *Environ. Res.* 156, 158–166.
- Ishido, M., et al., 2005. Alteration of gene expression of G protein-coupled receptors in endocrine disruptors-caused hyperactive rats. *Regul. Pept.* 126, 145–153.
- Kaun-Yu, L., et al., 2004. Suppression by phthalates of the calcium signaling of human nicotinic acetylcholine receptors in human neuroblastoma SH-SY5Y cells. *Toxicology* 200, 113–121.
- Kim, B.N., et al., 2009. Phthalates exposure and attention-deficit/hyperactivity disorder in school-age children. *Biol. Psychiatry* 66, 958–963.
- Kim, J.I., et al., 2017. The effects of maternal and children phthalate exposure on the neurocognitive function of 6-year-old children. *Environ. Res.* 156, 519–525.
- Kim, Y., et al., 2011. Prenatal exposure to phthalates and infant development at 6 months: prospective Mothers and Children's Environmental Health (MOCEH) study. *Environ. Health Perspect.* 119, 1495–1500.
- Koch, H.M., et al., 2005. New metabolites of di (2-ethylhexyl) phthalate (DEHP) in human urine and serum after single oral doses of deuterium-labelled DEHP. *Arch. Toxicol.* 79, 367–376.
- Komuro, H., Rakic, P., 1996. Intracellular Ca²⁺ fluctuations modulate the rate of neuronal migration. *Neuron* 17, 275–285.
- Lampen, A., et al., 2003. Teratogenic phthalate esters and metabolites activate the nuclear receptors PPARs and induce differentiation of F9 cells. *Toxicol. Appl. Pharmacol.* 188, 14–23.
- Lee, B.E., et al., 2014. Prenatal bisphenol A and birth outcomes: MOCEH (Mothers and Children's Environmental Health) study. *Int. J. Hyg. Environ. Health* 217, 328–334.
- Limbo, M.M., Joyce, D.P., 2011. Comparison of the ASQ and PEDS in screening for developmental delay in children presenting for primary care. *J. Dev. Behav. Pediatr.* 32, 499–511.
- Lin, C.H., et al., 2011. Activation of Trim17 by PPAR γ is involved in Di (2-ethylhexyl) phthalate (DEHP)-induced apoptosis on Neuro-2a cells. *Toxicol. Lett.* 206, 245–251.
- Lorz, P.M., et al., 2000. Phthalic Acid and Derivatives. *Ullmann's Encyclopedia of Industrial Chemistry*. Wiley-VCH Verlag GmbH & Co. KGaA, pp. 144–148.
- Mark H. Johnson, a.M. d.H., 2005. Developmental Cognitive Neuroscience. John Wiley & Sons, Ltd, West Sussex, UK.
- Miodovnik, A., et al., 2014. Developmental neurotoxicity of ortho-phthalate diesters: review of human and experimental evidence. *Neurotoxicology* 41, 112–122.
- Moore, R.W., et al., 2001. Abnormalities of sexual development in male rats with in utero and lactational exposure to the antiandrogenic plasticizer Di (2-ethylhexyl) phthalate. *Environ. Health Perspect.* 109, 229.
- Palleschi, S., et al., 2009. Di (2-ethylhexyl) phthalate stimulates Ca²⁺ entry, chemotaxis and ROS production in human granulocytes. *Toxicol. Lett.* 187, 52–57.
- Polanska, K., et al., 2014. Phthalate exposure and child development: the Polish Mother and Child Cohort Study. *Early Hum. Dev.* 90, 477–485.
- Porterfield, S.P., Hendrich, C.E., 1993. The role of thyroid hormones in prenatal and neonatal neurological development—current perspectives. *Endocr. Rev.* 14, 94–106.
- Rauh, V.A., Margolis, A.E., 2016. Research review: environmental exposures, neurodevelopment, and child mental health—new paradigms for the study of brain and behavioral effects. *J. Child Psychol. Psychiatry* 57, 775–793.
- Seybold, B.A., et al., 2012. Chronic reduction in inhibition reduces receptive field size in mouse auditory cortex. *Proc. Natl. Acad. Sci.* 109, 13829–13834.
- Silva, M.J., et al., 2004. Urinary levels of seven phthalate metabolites in the U.S. population from the National Health and Nutrition Examination Survey (NHANES) 1999–2000. *Environ. Health Perspect.* 112, 331–338.
- Smith, C., et al., 2011. Acute postnatal exposure to di (2-ethylhexyl) phthalate adversely impacts hippocampal development in the male rat. *Neuroscience* 193, 100–108.
- Suzuki, Y., et al., 2009. Exposure assessment of phthalate esters in Japanese pregnant women by using urinary metabolite analysis. *Environ. Health Perspect.* 117, 180.
- Swan, S.H., 2008. Environmental phthalate exposure in relation to reproductive outcomes and other health endpoints in humans. *Environ. Res.* 108, 177–184.

- Tang, F., et al., 2003. Spontaneous calcium transients in developing cortical neurons regulate axon outgrowth. *J. Neurosci.* 23, 927–936.
- Teitelbaum, S., et al., 2008. Temporal variability in urinary concentrations of phthalate metabolites, phytoestrogens and phenols among minority children in the United States. *Environ. Res.* 106, 257–269.
- Tellez-Rojo, M.M., et al., 2013. Prenatal urinary phthalate metabolites levels and neurodevelopment in children at two and three years of age. *Sci. Total Environ.* 461–462, 386–390.
- Tseng, I.L., et al., 2013. Phthalates induce neurotoxicity affecting locomotor and thermotactic behaviors and AFD neurons through oxidative stress in *Caenorhabditis elegans*. *PLoS One* 8, e82657.
- Ünür, T., Büyükgözü, A., 2012. Fetal and neonatal endocrine disruptors. *J. Clin. Res. Pediatr. Endocrinol.* 4, 51.
- Wada, K., et al., 2006. Peroxisome proliferator-activated receptor γ -mediated regulation of neural stem cell proliferation and differentiation. *J. Biol. Chem.* 281, 12673–12681.
- Wechsler, D., et al., 2012. Wechsler Preschool and Primary Scale of Intelligence—Fourth Edition.
- Wells, G., et al., 2013. The Newcastle-Ottawa Scale (NOS) for Assessing the Quality of Nonrandomized Studies in Meta-analyses 2017 Ottawa Hospital Research Institute.
- Whyatt, R.M., et al., 2012. Maternal prenatal urinary phthalate metabolite concentrations and child mental, psychomotor, and behavioral development at 3 years of age. *Environ. Health Perspect.* 120, 290–295.
- Williams, G.R., 2008. Neurodevelopmental and neurophysiological actions of thyroid hormone. *J. Neuroendocrinol.* 20, 784–794.
- Williams, J.H., Ross, L., 2007. Consequences of prenatal toxin exposure for mental health in children and adolescents. *Eur. Child. Adolesc. Psychiatry* 16, 243–253.
- Wu, W., et al., 2017. Exposure to phthalates in children aged 5–7 years: associations with thyroid function and insulin-like growth factors. *Sci. Tot. Environ.* 579, 950–956.
- Wu, Y., et al., 2014. Primary neuronal-astrocytic co-culture platform for neurotoxicity assessment of di-(2-ethylhexyl) phthalate. *J. Environ. Sci.* 26, 1145–1153.
- Xu, H., et al., 2013. Effects of di-n-butyl phthalate and diethyl phthalate on acetylcholinesterase activity and neurotoxicity related gene expression in embryonic zebrafish. *Bull. Environ. Contam. Toxicol.* 91, 635–639.
- Xu, Y., et al., 2007. Di-(2-ethylhexyl)-phthalate affects lipid profiling in fetal rat brain upon maternal exposure. *Arch. Toxicol.* 81, 57–62.